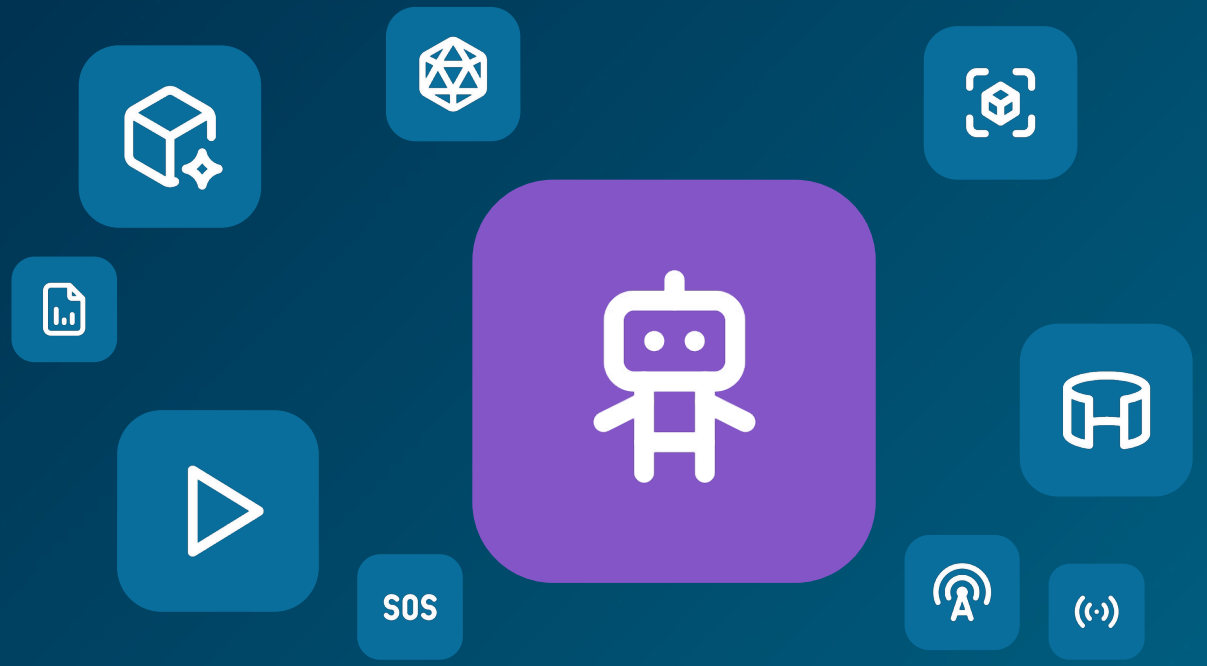
The Media Connectivity Association



OVERVIEW AND UPDATES · IMMERSIVE MEDIA

Conversational Avatar Communication

Real-time conversational avatars, animated from user to user

MPEG (ISO/IEC) 3GPP

NOVEMBER 2026 – www.5g-mag.com

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About the project

Real-time conversational avatars, animated from user to user

3GPP TR 26.813

Technical Report

Formats for representing and animating avatars, and recommendations for integrating them in 3GPP services. V20.0.0 is a Release 20 report; the studies are FS_AVATAR (Release 19) and FS_Avatar_Ph2_MED (Release 20, complete).

MPEG ARF

ISO/IEC 23090-39

The Avatar Representation Format: base avatar data model, container and animation stream.

3GPP TS 26.264

Technical Specification

IMS-based AR real-time communication, created in Release 18; Release 19 added avatar calls over the IMS data channel and the three animation modes. Release 20 (work plan of 28 September 2026): AvCall_Ph2-MED, planned finish 3 March 2027.

FIND OUT MORE Technology Specifications Reference Tools

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The avatar format: MPEG ARF

ISO/IEC 23090-39, as described in 3GPP TR 26.813

Container ISOBMFF or Zip

- ARF document
- Animation track (optional)

Base Avatar Format stores the avatar asset

- Skeletons
- Meshes
- Blendshapes
- Skins
- Landmarks
- Textures
- Nodes (joints)

Animation Stream Format live or stored

- Avatar Animation Units
- AAU header and payload
- configuration
- blendshape
- joint
- landmark
- texture

The base avatar and the animation are separate: a client loads the model once, then receives only the compact animation stream during the call.

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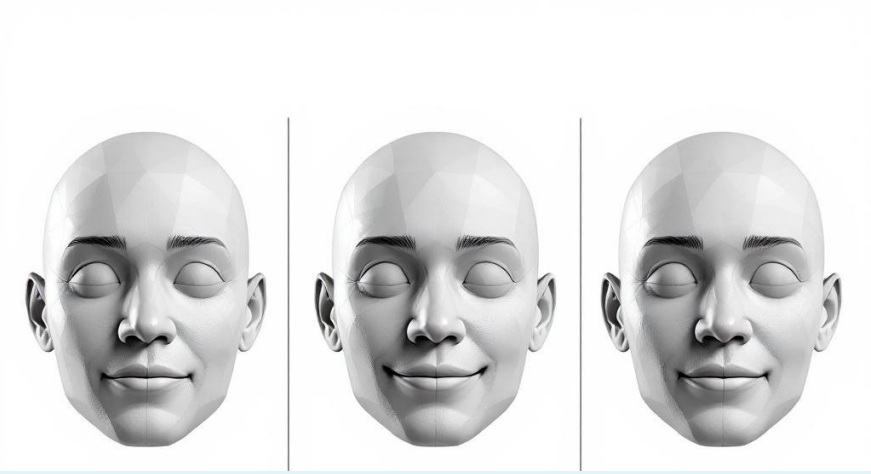
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Animating the avatar

Avatar Animation Units (AAU) of 3GPP TR 26.813

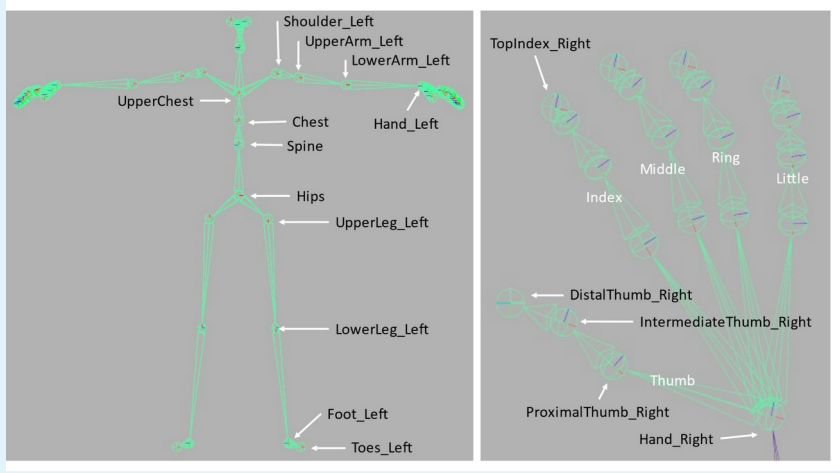
FIVE KINDS OF AAU

configuration | blendshape | joint | landmark | texture



Facial animation

Blend shape animation starts from the neutral expression and adds weighted contributions from predefined expressions.



Body and hand animation

Linear blend skinning moves the avatar surface with the skeleton, weighted per joint.

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Three avatar animation modes

Three avatar animation modes defined by 3GPP TS 26.264 for a call over the IMS data channel

SENDER-CENTRIC

The sender animates

"the sender UE animates and renders its avatar representation before sending it to the receiving UE as 2D video."

RECEIVER-CENTRIC

The receiver animates

"the receiving UE animates and renders the sender UE's avatar representation."

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NETWORK-CENTRIC

The network animates

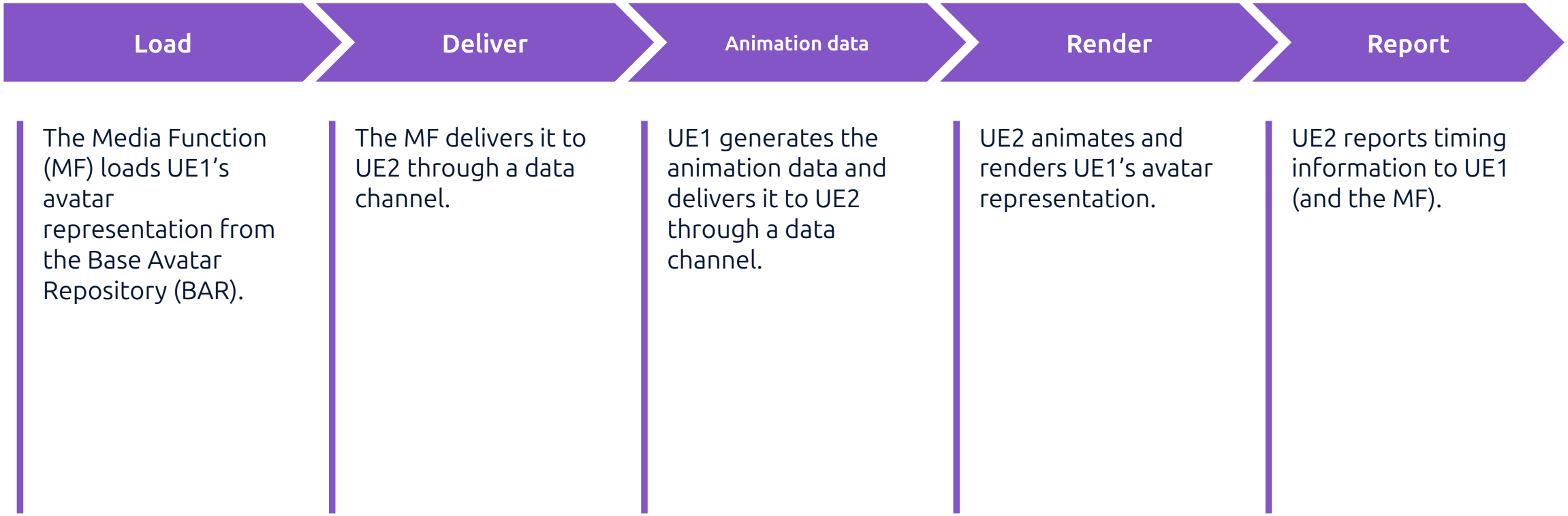
"the MF animates and renders the sender UE's avatar representation before sending it to the receiving UE as 2D video."

Base Avatar Repository: "stores the base avatar of a user and makes it available to other participants" (3GPP TR 26.813).

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A receiver-centric call

Avatar delivery and animation flow of 3GPP TS 26.264, receiver-centric mode



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Repositories

On main: a WebRTC call signalled through SWAP, carrying audio and avatar animation; IMS and network roles are not implemented

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
rt-avatar-call	the Unity call application: Android and Windows	Receiver-centric call, ARF format, Animation, SWAP signalling	v0.10.154 16 Oct 2026	GitHub
rt-avatar-content	reference MPEG ARF content for the call	ARF format	v0.10.154 16 Oct 2026	GitHub
rt-avatar-unity-tracker	a sample OpenXR tracker for ARF avatar animation in Unity; no code yet	Animation	none yet	GitHub
rt-avatar-bar	stores base avatar models as ARF containers, scoped to a call; client fetch pending	Base Avatar Repository	none yet	GitHub
rt-3gpp-swap	the 3GPP SWAP server and Node.js library (TS 26.113)	SWAP signalling	v0.9 16 Oct 2026	GitHub

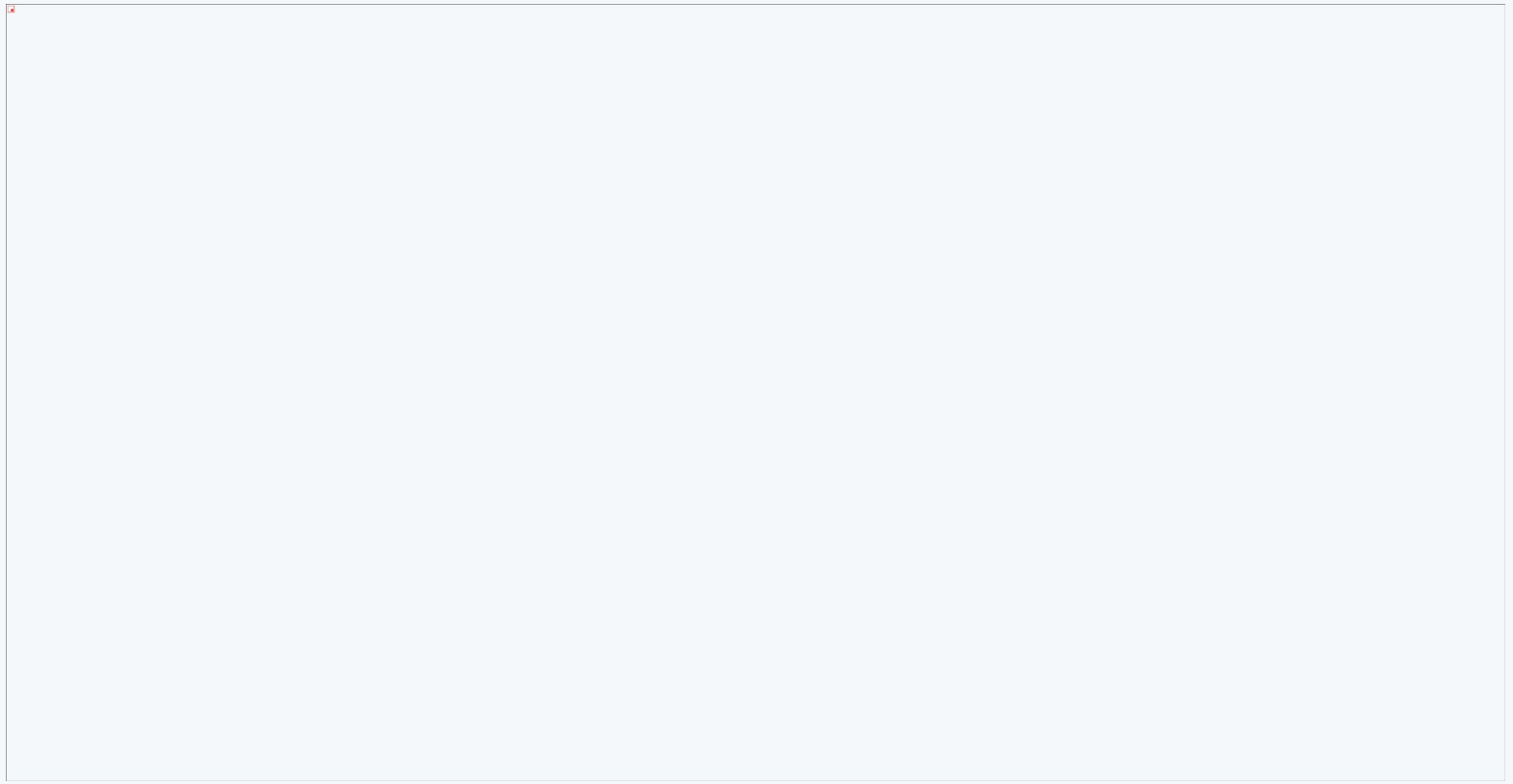
FIND OUT MORE **Technology** **Specifications** **Reference Tools**

● released ● no release yet

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Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them



PROVIDED BY THE REFERENCE TOOLS

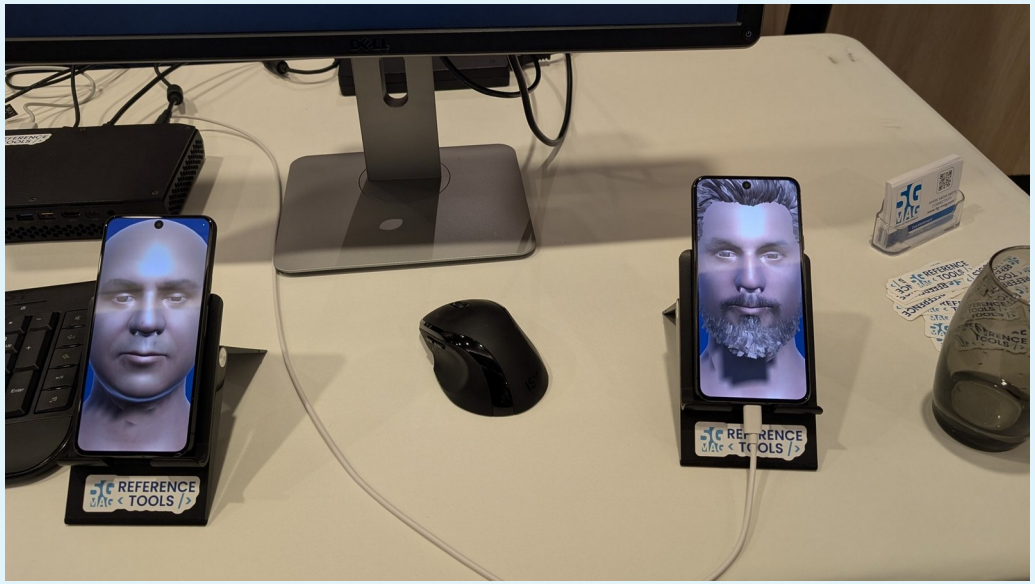
1. **rt-avatar-call**
2. **rt-avatar-content**
3. **MediaPipe**
   dependency, used by 1
4. **UnityARF**
   dependency, used by 1

Boxes without a badge are not mapped to a repository.

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See it and try it

Photos of the project's demo, an avatar call on two phones



MPEG meeting, Geneva, 14 July 2026



IBC 2026, 13 September 2026

CONTRIBUTORS MOTION SPELL Qualcomm interdigital.

WHERE WE STAND

Conversational Avatars with MPEG ARF

BROUGHT TO STUDY	ON-GOING	OUTCOMES
Avatar calls phase 2	Gaussian splatting avatars	Initial Avatar Call Client
Body animation support in the avatar call client	Animation generation on Android XR	Initial implementation of SWAP protocol
	Base Avatar Repository	Avatar content for the call client
	Android Avatar Call Client Application with enhanced UX	

RELATION WITH SDOS AND INDUSTRY FORA MPEG (ISO/IEC) 3GPP

MPEG Avatar part of contribution m78825 at MPEG#156 (Hangzhou, October 2026): a receiver-centric avatar call on two phones

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